2.8 Servo UNO

Servo

Overview

Servo is a type of geared motor that can only rotate 180 degrees. It is controlled by sending electrical pulses from your UNO board. These pulses tell the servo what position it should move to. The Servo has three wires, of which the brown one is the ground wire and should be connected to the GND port of UNO, the red one is the power wire and should be connected to the 3.3V port, and the orange one is the signal wire and should be connected to the GPIO port.

Component Required:

(1) x Elegoo Uno R3

(1) x Servo (SG90)

(3) x M-M wires (Male to Male jumper wires)

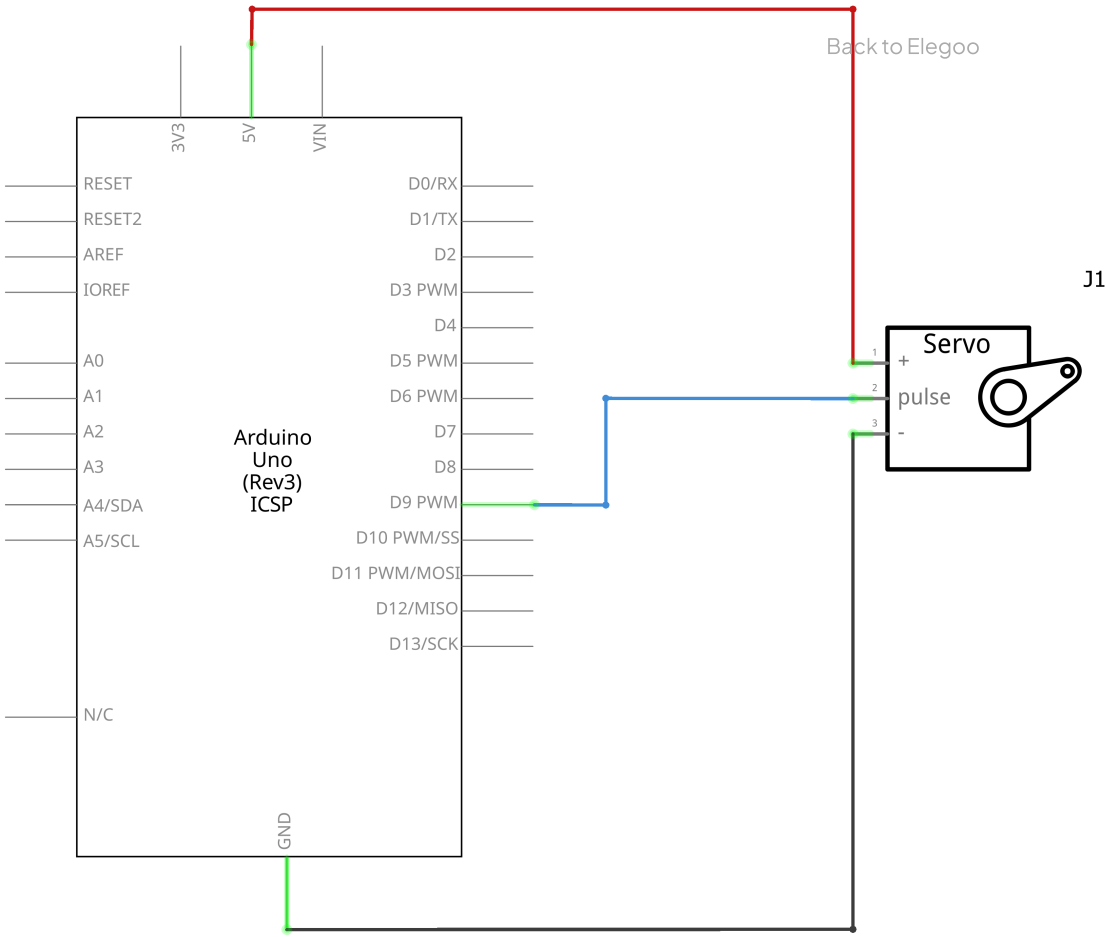
Component Introduction

SG90

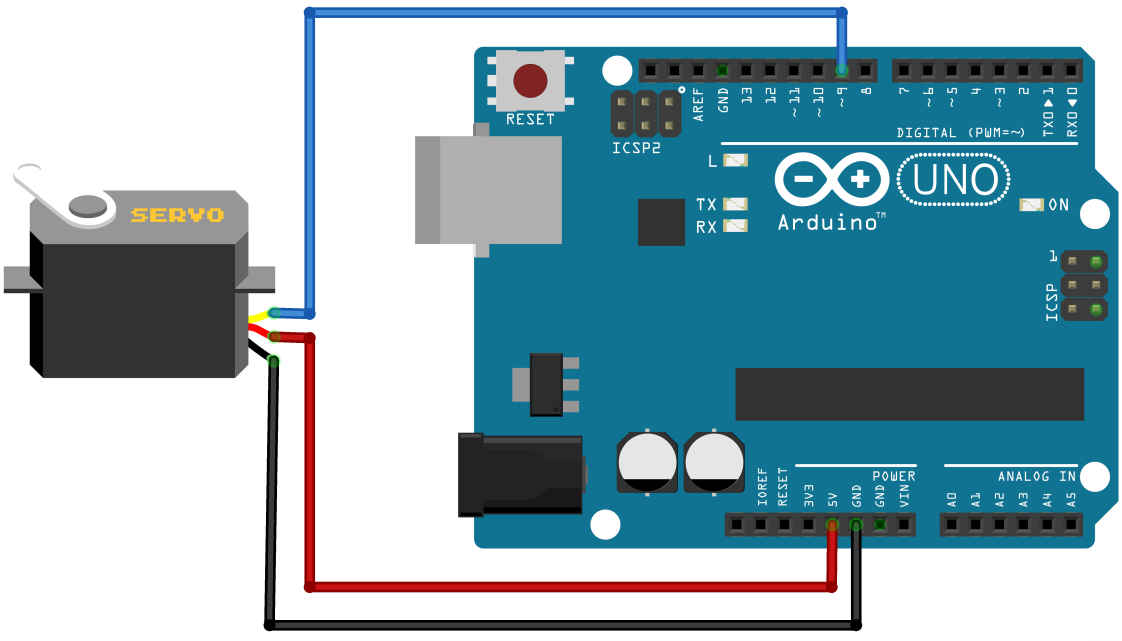
·Universal for JR and FP connector
·Cable length : 25cm
·No load; Operating speed: 0.12 sec / 60 degree (4.8V), 0.10 sec / 60 degree (6.0V)
·Stall torque (4.8V): 1.6kg/cm
·Temperature : -30~60'C
·Dead band width: 5us
·Working voltage: 3.5~6V
·Dimension : 1.26 in x 1.18 in x 0.47 in (3.2 cm x 3 cm x 1.2 cm)
·Weight : 4.73 oz (134 g)



Connection Schematic



Wiring diagram



Code

You can click the blue text link to download the program file to your local device, and double-click the file to open it after the download is complete. Please note: Before opening the file, ensure that you have installed the Arduino IDE development environment and completed the installation of relevant components such as the board support package and driver corresponding to the uno development board. If you have any questions about this operation process, you can refer to the "part 1" chapter of the document for detailed guidance

Before you can run this, make sure that you have installed the **Servo** library or re-install it, if necessary. Otherwise, your code won't work.

```
#include <Servo.h>

/* After including the corresponding libraries,
we can use the "class" like "Servo" created by the developer for us.
We can use the functions and variables created in the libraries by creating
objects like the following "myservo" to refer to the members in ".".*/

Servo myservo;
//it created an object called myservo.
/* you can see Servo as a complex date type(Including functions and various data types)
and see myservo as variables. */
```

Library and Object Configuration:

- `#include <Servo.h>` : Includes the Servo library ()
- `Servo myservo` : Creates a servo object named "myservo" that allows control of a single servo motor

Object-Oriented Programming (OOP) Concepts:

- `Servo` is a class that encapsulates servo control functionality
- `myservo` is an instance (object) of the Servo class
- The object allows access to member functions like `attach()` and `write()`

```
void setup(){
/*"attach" and "write" are both functions,
and they are members contained in the complex structure of "Servo".
We can only use them if we create the object "myservo" for the complex structure of "Servo".
*/
myservo.attach(9);//connect pin 9 with the control line(the middle line of Servo)
myservo.write(90);// move servos to center position -> 90°
}
```

setup() Function: Initializes the system and configures the servo motor.

- `myservo.attach(9)` : Attaches the servo object to pin 9, which connects to the servo's control wire
- `myservo.write(90)` : Initializes the servo to the center position (90°)

Servo Wiring:

- Most servos have three wires:
  - **Control wire** (usually yellow or orange): Connects to D9
  - **Power wire** (usually red): Connects to 3.3V or 5V power supply
  - **Ground wire** (usually brown or black): Connects to GND

Servo Initialization:

- Setting the servo to 90° at startup ensures it begins at a known position
- This prevents unexpected movement when the code starts running

```
void loop(){
myservo.write(90);// move servos to center position -> 90°
delay(1000);
myservo.write(60);// move servos to center position -> 60°
delay(1000);
myservo.write(90);// move servos to center position -> 90°
delay(1000);
myservo.write(150);// move servos to center position -> 150°
delay(1000);
}
```

loop() Function Overview: The main loop that controls the servo motor's movement sequence.

Servo Movement Sequence:

1. **Center Position (90°):**

```
myservo.write(90);// move servos to center position -> 90°
delay(1000);
```

   - Moves the servo to the center position (90°)
   - Pauses for 1000 milliseconds (1 second)

2. **Left Position (60°):**

```
myservo.write(60);// move servos to center position -> 60°
delay(1000);
```

   - Moves the servo to the left position (60°)
   - Pauses for 1 second

3. **Center Position (90°) - Return:**

```
myservo.write(90);// move servos to center position -> 90°
delay(1000);
```

   - Returns the servo to the center position (90°)
   - Pauses for 1 second

4. **Right Position (150°):**

```
myservo.write(150);// move servos to center position -> 150°
delay(1000);
```

   - Moves the servo to the right position (150°)
   - Pauses for 1 second before repeating the loop

Servo Movement Pattern:

Sequence Step	Angle	Duration	Movement Description
1	90°	1 second	Start at center
2	60°	1 second	Move left
3	90°	1 second	Return to center
4	150°	1 second	Move right

Servo Motor Fundamentals:

- **PWM Control**: Servos use Pulse Width Modulation (PWM) signals to determine position
- **Angle Range**: Most servos rotate between 0° and 180°
- **Control Signal**:
  - 1ms pulse → 0° position
  - 1.5ms pulse → 90° position (center)
  - 2ms pulse → 180° position

Troubleshooting:

- **Servo not moving**: Check power supply, wiring, and that the servo is compatible with the library
- **Jittering**: Ensure stable power supply and proper grounding
- **Incorrect angle**: Verify the servo's actual rotation range matches the code's angle assumptions
- **Limited movement**: Some servos have restricted rotation ranges

Servo Library Functions:

- attach(pin) : Attaches the servo to a GPIO pin
- write(angle) : Moves the servo to the specified angle (0–180°)
- writeMicroseconds(us) : Controls the servo using microsecond pulse widths (alternative to angle control)
- read() : Returns the current servo position (in degrees)
- attached() : Returns true if the servo is attached to a pin
- detach() : Detaches the servo from its pin, releasing it for other uses

←Tilt Ball

Ultrasonic sensor→

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